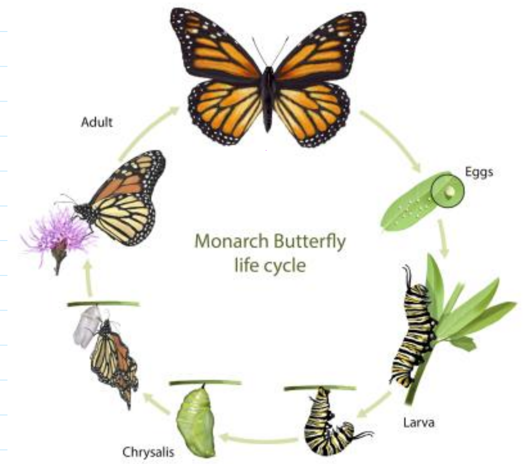
Repetition

05 August 2025 06:09

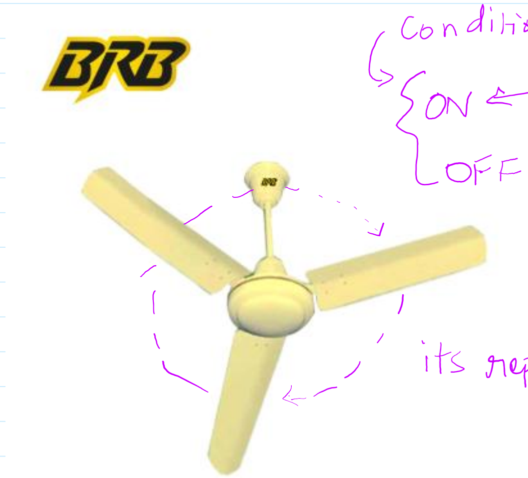
← representing time.

hr min sec
1 hr 1 1 ... 60
2 1 ... 60
: :
60



Iterative process occurring in nature.

Fan.



condition.
ON ← electric supply is on.
OFF ← cut electric supply.

& produces air

its repeating its motion.

Repetition
Often we repeat a task, for example payment of electricity bill which is done every month. In in the above figure. It shows the life cycle of butterfly that involves four stages. That is, a butterfly lay eggs, turns into Caterpillar, becomes a pupa, and finally matures as a butterfly. The cycle starts again with laying eggs of the butterfly.
This kind of repetition is also called iteration. Repetition of a set of statements in a program is made possible using looping constructs.

```python
n = int(input('Enter any terms: '))
start = 1
while start <= n:
    print(start)
    start += 1
```

condition

Repeating [till condion is true]

Keyword in Python.

Syntax:

while <condition>:
    statement(s).

The while loop execute statement(s) written under its block till the condition is True.

WAP to input number of term and print all the even numbers from n to 'n'(n is inclusive if n is even)

Write a program to input number of values. And the user will input. Positive values and negative values. And the program outputs the sum of positive values and the negative values. For example:

Enter number of values you have: 5
  Enter any number: 10
  Enter any number: -1
  Enter any number: -7
  Enter any number: -3
  Enter any number: 5
Output:
  Sum of positive values are: 15
  Sum of negative values are: -11

Home work:

1. Write separate programs in Python to display the first N terms of the following series:
   (a) 1, 4, 9, 16, …
   (b) 1, 2, 4, 7, 11, …
   (c) 3, 6, 9, 12, …
   (d) 4, 8, 16, 32,…
   (e) 1.5, 3.0, 4.5, 6.0,…

(a) 1, 4, 9, 16, …
Input: Enter number of terms: 7
Output: 1, 4, 9, 16, 25, 36, 49

```python
# 1. Write separate programs in Python to display the first N terms of the following  series:
# Output: 1, 2, 4, 7, 11, …
# 1, 1+1=2, 2+2=4, 4+3=7, 7+4=11, ... so on
n = int(input('Enter number of terms: '))
start = 0
t = 1
while start < n - 1: #n=5
    t += start       # 1, 2, 4, 7, 11
    print(t,', ',sep='',end='') #1, 2, 4, 7,
    start += 1       # 1, 2, 3, 4, 5
t += start # 11
print(t) #1, 2, 4, 7, 11
```

Output:

```
Enter number of terms: 7
1, 2, 4, 7, 11, 16, 22
```

What is a nested loop?

A nested **while** loop means one **while** loop inside another while loop.

- The outer loop controls the number of times the inner loop runs.
- For every single execution of the outer loop, the inner loop runs completely.

Syntax:

```
while condition1:   #Outer loop
  #Outer loop code
   while condition2:  # Inner loop
     #Inner loop code
  #Outer loop code
```

```python
start = 1
N = 20
while start <= N:
    t = 1
    print('Table of', start)
    while t <= 10:
        p = start * t
        print(start,'*',t,'=',p)
        t +=1
    print()
    start += 1
```

start = ~~1~~ 2

2 <= 20

t = 1 ~~2~~ 3

p = ~~2~~ 4 6

Output :

Table of 1
1 * 1 = 1
1 * 2 = 2
1 * 3 = 3
:
1 * 10 = 10

Table of 2
2 * 1 = 2
2 * 2 = 4
start → 2 * 3 = 6 ← P
t

Output:

```
Table of 1
1 * 1 = 1
1 * 2 = 2
1 * 3 = 3
```

1 * 4 = 4
1 * 5 = 5
1 * 6 = 6
1 * 7 = 7
1 * 8 = 8
1 * 9 = 9
1 * 10 = 10

Table of 2
2 * 1 = 2
2 * 2 = 4
2 * 3 = 6
2 * 4 = 8
2 * 5 = 10
2 * 6 = 12
2 * 7 = 14
2 * 8 = 16
2 * 9 = 18
2 * 10 = 20

Table of 3
3 * 1 = 3
3 * 2 = 6
3 * 3 = 9
3 * 4 = 12
3 * 5 = 15
3 * 6 = 18
3 * 7 = 21
3 * 8 = 24
3 * 9 = 27
3 * 10 = 30

Table of 4
4 * 1 = 4
4 * 2 = 8
4 * 3 = 12
4 * 4 = 16
4 * 5 = 20
4 * 6 = 24
4 * 7 = 28
4 * 8 = 32
4 * 9 = 36
4 * 10 = 40

Table of 5
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50

Table of 6
6 * 1 = 6
6 * 2 = 12
6 * 3 = 18
6 * 4 = 24
6 * 5 = 30

6 * 6 = 36
6 * 7 = 42
6 * 8 = 48
6 * 9 = 54
6 * 10 = 60

Table of 7
7 * 1 = 7
7 * 2 = 14
7 * 3 = 21
7 * 4 = 28
7 * 5 = 35
7 * 6 = 42
7 * 7 = 49
7 * 8 = 56
7 * 9 = 63
7 * 10 = 70

Table of 8
8 * 1 = 8
8 * 2 = 16
8 * 3 = 24
8 * 4 = 32
8 * 5 = 40
8 * 6 = 48
8 * 7 = 56
8 * 8 = 64
8 * 9 = 72
8 * 10 = 80

Table of 9
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90

Table of 10
10 * 1 = 10
10 * 2 = 20
10 * 3 = 30
10 * 4 = 40
10 * 5 = 50
10 * 6 = 60
10 * 7 = 70
10 * 8 = 80
10 * 9 = 90
10 * 10 = 100

Table of 11
11 * 1 = 11
11 * 2 = 22
11 * 3 = 33
11 * 4 = 44
11 * 5 = 55
11 * 6 = 66
11 * 7 = 77

11 * 8 = 88
11 * 9 = 99
11 * 10 = 110

Table of 12
12 * 1 = 12
12 * 2 = 24
12 * 3 = 36
12 * 4 = 48
12 * 5 = 60
12 * 6 = 72
12 * 7 = 84
12 * 8 = 96
12 * 9 = 108
12 * 10 = 120

Table of 13
13 * 1 = 13
13 * 2 = 26
13 * 3 = 39
13 * 4 = 52
13 * 5 = 65
13 * 6 = 78
13 * 7 = 91
13 * 8 = 104
13 * 9 = 117
13 * 10 = 130

Table of 14
14 * 1 = 14
14 * 2 = 28
14 * 3 = 42
14 * 4 = 56
14 * 5 = 70
14 * 6 = 84
14 * 7 = 98
14 * 8 = 112
14 * 9 = 126
14 * 10 = 140

Table of 15
15 * 1 = 15
15 * 2 = 30
15 * 3 = 45
15 * 4 = 60
15 * 5 = 75
15 * 6 = 90
15 * 7 = 105
15 * 8 = 120
15 * 9 = 135
15 * 10 = 150

Table of 16
16 * 1 = 16
16 * 2 = 32
16 * 3 = 48
16 * 4 = 64
16 * 5 = 80
16 * 6 = 96
16 * 7 = 112
16 * 8 = 128
16 * 9 = 144

```
16 * 10 = 160

Table of 17
17 * 1 = 17
17 * 2 = 34
17 * 3 = 51
17 * 4 = 68
17 * 5 = 85
17 * 6 = 102
17 * 7 = 119
17 * 8 = 136
17 * 9 = 153
17 * 10 = 170

Table of 18
18 * 1 = 18
18 * 2 = 36
18 * 3 = 54
18 * 4 = 72
18 * 5 = 90
18 * 6 = 108
18 * 7 = 126
18 * 8 = 144
18 * 9 = 162
18 * 10 = 180

Table of 19
19 * 1 = 19
19 * 2 = 38
19 * 3 = 57
19 * 4 = 76
19 * 5 = 95
19 * 6 = 114
19 * 7 = 133
19 * 8 = 152
19 * 9 = 171
19 * 10 = 190

Table of 20
20 * 1 = 20
20 * 2 = 40
20 * 3 = 60
20 * 4 = 80
20 * 5 = 100
20 * 6 = 120
20 * 7 = 140
20 * 8 = 160
20 * 9 = 180
20 * 10 = 200
```

```
start = 1
N = 20
print('Row-wise multiplication table: ')
while start <= N:
    t = 1
    print('Table of', start)
    while t < 10:
        p = start * t
        print(p,end=', ')
        t +=1
    print((start*t))
```

```
    start += 1
```

```
Output:
Row-wise multiplication table:
Table of 1
1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Table of 2
2, 4, 6, 8, 10, 12, 14, 16, 18, 20
Table of 3
3, 6, 9, 12, 15, 18, 21, 24, 27, 30
Table of 4
4, 8, 12, 16, 20, 24, 28, 32, 36, 40
Table of 5
5, 10, 15, 20, 25, 30, 35, 40, 45, 50
Table of 6
6, 12, 18, 24, 30, 36, 42, 48, 54, 60
Table of 7
7, 14, 21, 28, 35, 42, 49, 56, 63, 70
Table of 8
8, 16, 24, 32, 40, 48, 56, 64, 72, 80
Table of 9
9, 18, 27, 36, 45, 54, 63, 72, 81, 90
Table of 10
10, 20, 30, 40, 50, 60, 70, 80, 90, 100
Table of 11
11, 22, 33, 44, 55, 66, 77, 88, 99, 110
Table of 12
12, 24, 36, 48, 60, 72, 84, 96, 108, 120
Table of 13
13, 26, 39, 52, 65, 78, 91, 104, 117, 130
Table of 14
14, 28, 42, 56, 70, 84, 98, 112, 126, 140
Table of 15
15, 30, 45, 60, 75, 90, 105, 120, 135, 150
Table of 16
16, 32, 48, 64, 80, 96, 112, 128, 144, 160
Table of 17
17, 34, 51, 68, 85, 102, 119, 136, 153, 170
Table of 18
18, 36, 54, 72, 90, 108, 126, 144, 162, 180
Table of 19
19, 38, 57, 76, 95, 114, 133, 152, 171, 190
Table of 20
20, 40, 60, 80, 100, 120, 140, 160, 180, 200
```

Write separate program in Python to input number of rows and print the following patterns:

```
Enter number of rows: 5
Output:
1.
#
# #
# # #
# # # #
# # # # #

2.
# # # # #
# # # #
```

```
# # #
# #
#
```

```python
r=int(input("Enter the number of rows: "))
start=1
while start<=r:
    instart=1
    while instart<=start:
        print('#',end=' ')
        instart+=1
    print()   #changing the line
    start+=1
```

False

r = 5
start = ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6
instart = ~~1~~ ~~2~~ ~~1~~ ~~2~~ ~~3~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6

Output:

```
#_
#_#_
#_#_#_
#_#_#_#
#_#_#_#_#
_
```

3

```
        #
      # #
    # # #
  # # # #
# # # # #
```

Home work:

Write separate programs in Python to display the first N terms of the following series:

1^3-1 2^3-1 3^3-1

(f) 0, 7, 26, ...

(g) 1, 9, 25, 49, ...

(h) 4, 16, 36, 64, ...

(i) 0, 3, 8, 15, ...

(j) 24, 99, 224, 399, ...

(k) 2, 5, 10, 17, ...

2. Write a program to input any ~~50~~ 10 numbers (including positive and negative). Perform the following tasks:
   (a) Count and display the positive numbers
   (b) Count and display the negative numbers
   (c) Display the sum of positive numbers
   (d) Display the sum of negative numbers
3. Write a program to calculate and display the sum of all odd numbers and even numbers between a range of numbers from m to n (both inclusive) where m < n. Input m and n (where m<n).
4. Write a program to enter any 50 numbers and check whether they are divisible by 5 or not. If divisible then perform the following tasks:
   (a) Display all the numbers ending with the digit 5.
   (b) Count and display the numbers ending with 0 (zero).

Added home work:

5. Write a program to display all the numbers between m and n input from the keyboard (where m<n, m>0, n>0), check and display the numbers that are perfect square. e.g. 25, 36, 49, --------- are said to be perfect square numbers. [A number is said to be a perfect square if its square root is an integer number.]
6. Write a program to display all the 'Buzz Numbers' between p and q (where p<q). A 'Buzz Number' is the number which ends with 7 or is divisible by 7.
7. Write a program to input marks in English, Maths and Science of 40 students who have passed ICSE Examination 2014. Now, perform the following tasks:
   (a) Find and display the number of students, who have secured 95% or more in aggregate.
   (b) Find and display the number of students, who have secured 90% or more in English, Maths and Science.

Write a program to input number of terms and print the series along with sum:

(a) $1 + 4 + 9 + \ldots\ldots$

(b) $1 + \frac{1}{2} + \frac{1}{3} + \ldots\ldots$

(c) $1 + \frac{1}{3} + \frac{1}{5} + \ldots\ldots$

Home work

(d) $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \ldots\ldots$

(e) $2 - 4 + 6 - 8 + \ldots\ldots$

(f) $(1*2) + (2*3) + \ldots\ldots$

Home work

8. Write a program to input a number and display the new number after reversing the digits of the original number. The program also displays the absolute difference between the original number and the reversed number.

Sample Input: 194

Sample Output: 491

Absolute Difference= 297

9. The Greatest Common Divisor (GCD) of two integers is calculated by the continued division method. Divide the larger number by the smaller, the remainder then divides the previous divisor. The process repeats unless the remainder reaches zero. The last divisor results in GCD. [ICSE-2009]
Write a program to accept two numbers. Find and display their GCD by using the technique discussed above.

Sample Input: 45, 20

Sample Output: GCD=5